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NAAC GRADE A+

Student Name: Sahil Gupta **UID: 25MCI10266**
Branch: MCA(AI/ML) **Section/Group: 25MAM-1 A**
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Subject Name: TECHINCAL SKILLS **Subject Code:**

WORKSHEET 6

AIM: Learn how to create, query, and manage views in SQL to simplify database queries and provide a layer of abstraction for end-users.

S/W Requirement: Oracle Database Express Edition and pgAdmin

OBJECTIVES:

- Data Abstraction: To understand how to hide complex table joins and calculations behind a simple virtual table interface.
- Enhanced Security: To learn how to restrict user access to sensitive columns by providing views instead of direct table access.
- Query Simplification: To master the creation of views that pre-join multiple tables, making reporting easier for non-technical users.
- View Management: To understand the syntax for creating, altering, and dropping views, as well as the naming conventions required for efficient data access.

Practical / Experiment Steps

Table creation:

```
CREATE TABLE Department (
   dept_id SERIAL PRIMARY KEY,
   dept_name VARCHAR(50)
);

Create table employee(
    Emp_id serial primary key,
    Emp_name varchar (50),
    Salary numeric(10,2),
    Status varchar(20),
    Dept_id int references department(dept_id)
);

INSERT INTO Department (dept_name) VALUES
('HR'),
('IT'),
('Finance'),
```

```
('Sales');

Insert into employee(emp_name, salary, status, dept_id) values
('Kundan',40000,'Active', 1),
('Ranjan', 10000,'Inactive',2),
('Sahil', 43000,'Active', 2),
('Ram', 25000,'Active',3),
('Laxmi', 60000, 'Active', 4);
```

Output:

	dept_id [PK] integer	dept_name character varying (50)
1	1	HR
2	2	IT
3	3	Finance
4	4	Sales

Employee table :

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)	status character varying (20)	dept_id integer
1	1	Kundan	40000.00	Active	1
2	2	Ranjan	10000.00	Inactive	2
3	3	Sahil	43000.00	Active	2
4	4	Ram	25000.00	Active	3
5	5	Laxmi	60000.00	Active	4

Step 1: Creating a Simple View for Data Filtering

- Implementing a view to provide a quick list of active employees without exposing the entire table structure.

```
CREATE VIEW Active_Employees AS

SELECT emp_id, emp_name, salary, dept_id

FROM Employee

WHERE status = 'Active';
```

Output:

	emp_id integer	emp_name character varying (50)	salary numeric (10,2)	dept_id integer
1	1	Kundan	40000.00	1
2	3	Sahil	43000.00	2
3	4	Ram	25000.00	3
4	5	Laxmi	60000.00	4

Step 2: Creating a View for Joining Multiple Tables

- Simplifying the retrieval of data distributed across Employees and Departments tables.

```
CREATE VIEW Employee_Department_View AS
SELECT
   e.emp_id,
   e.emp_name,
   e.salary,
   e.status,
   d.dept_name
FROM Employee e
JOIN Department d
   ON e.dept_id = d.dept_id;
```

Output:

	emp_id integer	emp_name character varying (50)	salary numeric (10,2)	status character varying (20)	dept_name character varying (50)
1	1	Kundan	40000.00	Active	HR
2	2	Ranjan	10000.00	Inactive	IT
3	3	Sahil	43000.00	Active	IT
4	4	Ram	25000.00	Active	Finance
5	5	Laxmi	60000.00	Active	Sales

Step 3: Advanced Summarization View

- Creating a view to provide department-level statistics automatically

```
CREATE VIEW Department_Salary_Summary AS
SELECT
   d.dept_id,
   d.dept_name,
   COUNT(e.emp_id) AS total_employees,
   COUNT(CASE WHEN e.status = 'Active' THEN 1 END) AS active_employees,
   SUM(e.salary) AS total_salary,
```

```
    AVG(e.salary) AS avg_salary,
    MAX(e.salary) AS highest_salary,
    MIN(e.salary) AS lowest_salary
FROM Department d
LEFT JOIN Employee e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;
```

Output:

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	dept_id integer	dept_name character varying (50)	total_employees bigint	active_employees bigint	total_salary numeric	avg_salary numeric	highest_salary numeric	lowest_salary numeric
1	4	Sales	1	1	60000.00	60000.000000000000	60000.00	60000.00
2	2	IT	2	1	53000.00	26500.000000000000	43000.00	10000.00
3	3	Finance	1	1	25000.00	25000.000000000000	25000.00	25000.00
4	1	HR	1	1	40000.00	40000.000000000000	40000.00	40000.00

Outcomes:

- **Abstraction Proficiency:** Students will be able to create and query views to simplify efficient data access and abstraction.
- **Security Implementation:** Students will understand how to use views for data masking and providing restricted access to sensitive information.
- **Syntactic Accuracy:** Students will demonstrate the correct syntax for creating and querying views, ensuring logical clarity in naming conventions.
- **Real-world Application:** Students will be able to design views for practical domains like Library Management Systems or Payroll Systems to demonstrate functionality.